

# Skills Summary

## Composing Piecewise and Nonlinear Functions

Use this reference while completing your worksheet or when studying.

**Definition.**  $(f \circ g)(x) = f(g(x))$  — the **inner** function  $g$  acts first, and its output becomes the input of  $f$ .

**Two rules that cause most errors.**

- Order matters:  $f \circ g$  and  $g \circ f$  are different functions.
- Domain: the input must be legal for  $g$ , and  $g(x)$  must be legal for  $f$ . Simplifying can hide a restriction, so read the domain off the composition before simplifying.

Examples on this card use  $a(x) = 3x - 1$ ,  $b(x) = x^2 + 2$ , and  $w(x) = \begin{cases} -x, & x < 0 \\ x + 6, & x \geq 0 \end{cases}$

### 1. Evaluating a composition at a number

Work **inside out**. Compute the inner value, write it down, then substitute that single number into the outer function. Never substitute into both at once.

**Example:** Find  $(a \circ b)(2)$ .

**Step 1.** The composition names  $b$  as the inner function, so the only legal first move is to evaluate  $b$  at 2 and carry that output outward.

**Step 2.**  $b(2) = 2^2 + 2 = 6$ , and then  $a(6) = 3(6) - 1$ .

$$(a \circ b)(2) = \mathbf{17}$$

**Try it:** Find  $(b \circ a)(2)$  — the same two functions, other order.

check: 27

### 2. Simplifying a composition into one rule

Substitute the whole inner *rule* into the outer function's variable slot, keeping it in brackets, then expand:  $a(b(x)) = 3(b(x)) - 1$ . Expand squares as  $(mx + c)^2 = m^2x^2 + 2mcx + c^2$  — never as  $m^2x^2 + c^2$ .

**Example:** Write  $(a \circ b)(x)$  as one simplified rule.

**Step 1.** A rule, not a number, is wanted, so the inner rule goes into the slot where  $a$ 's variable stands and nothing is evaluated yet.

**Step 2.**  $a(b(x)) = 3(x^2 + 2) - 1 = 3x^2 + 6 - 1$ .

$$\mathbf{3x^2 + 5} \quad (\text{check at } x = 2 : 12 + 5 = 17)$$

**Try it:** Write  $(b \circ a)(x)$  as one simplified rule.

check:  $x^3 + 9x - 6$

### 3. Piecewise outer function: pick the piece

A piecewise function tests **its own input**. In  $w(g(x))$  that input is  $g(x)$ , *not*  $x$  — so compute the inner value first, compare *that* with the break point, and only then choose a rule. For a piecewise rule in  $x$ , translate the break:  $w$  breaks at 0, so  $w(a(x))$  breaks where  $a(x) = 0$ .

**Example:** Find  $w(a(0))$ .

**Step 1.** The piece is chosen by what reaches  $w$ , so evaluate the inner function before looking at either condition.

**Step 2.**  $a(0) = 3(0) - 1 = -1$ . Since  $-1 < 0$ ,  $w$  uses the  $-x$  piece, giving  $-(-1)$ .

$$w(a(0)) = 1$$

**Try it:** Find  $w(a(2))$ . (Check the sign of  $a(2)$  before choosing.)

check: 11

### 4. Solving a composition equation

Simplify the composition to one rule, set it equal to the target, and solve. Quadratic compositions usually have **two** solutions — take both roots.

With a piecewise outer function, solve *each* piece and then test every candidate against the condition that selects that piece; a candidate outside its own region is rejected.

**Example:** Solve  $(a \circ b)(x) = 32$ .

**Step 1.** One rule for the composition already exists from section 2, so the equation can be solved directly instead of unwinding two functions.

**Step 2.**  $3x^2 + 5 = 32$  gives  $3x^2 = 27$ , so  $x^2 = 9$ , which has two real roots.

$$x = -3 \text{ or } x = 3$$

**Try it:** Solve  $(b \circ a)(x) = 3$ , using the rule from section 2.

check:  $\frac{x}{2} = x$  to  $0 = x$